# Scalability & Future Plan

| Field | Value |
|---|---|
| Date | 29 October 2026 |
| Team ID | [ENTER ACTUAL TEAM ID] |
| Project Name | EV Market Intelligence Dashboard |
| Maximum Marks | 1 Mark |

## Scalability & Future Plan

The current solution is a CSV-backed Tableau Public dashboard. The upgrades below are proposals, not features claimed to be already implemented.

### Current System Limitations

| S.No | Limitation | Impact | Priority to Address |
|---|---|---|---|
| 1 | CSV is a static input and requires manual replacement to refresh data. | New data is not automatically reflected in the published dashboard. | Medium |
| 2 | Dense scatter plot and multiple worksheets can make the dashboard slower or crowded in a browser. | Users may experience slower rendering or need to scroll/zoom. | High |
| 3 | Data quality and representativeness are not independently verified. | Charts may be misinterpreted as real-world market facts. | High |

### Scalability Plan

| S.No | Scalability Aspect | Current State | Proposed Upgrade / Solution |
|---|---|---|---|
| 1 | User Load | Shared Tableau Public dashboard; usage depends on the hosted service. | Keep visuals lightweight, limit unnecessary marks and test the published view on typical devices. |
| 2 | Data Storage | One CSV file (100,000 records; 37 fields). | For larger or frequently refreshed data, assess a managed database or curated extract and define a refresh process. |
| 3 | Performance | Browser-rendered dashboard with several worksheets and a dense scatter plot. | Reduce mark density where appropriate, avoid unnecessary labels, use filters carefully and test render behaviour. |
| 4 | Security | Published Tableau Public content is intended to be publicly accessible. | Publish only non-confidential data; use an approved private platform if access control is required. |

### Future Roadmap

| Phase | Planned Feature / Enhancement | Target Timeline | Expected Impact |
|---|---|---|---|
| Phase 1 | Improve dashboard fit, labels, tooltips and layout; validate public URL. | Next iteration | More readable presentation and fewer clipped/overlapping views. |
| Phase 2 | Add verified KPI definitions and targeted filters/actions. | After review feedback | Faster exploration of stakeholder questions. |
| Phase 3 | Validate data quality and compare results with trustworthy external sources. | When suitable data is available | More credible interpretation of market and sustainability metrics. |
| Phase 4 | Assess automated refresh and a scalable data source if recurring updates are required. | Future enhancement | Less manual maintenance and better support for larger datasets. |